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> VerifyLocalTaylorBounds := proc()
    local CoefficientInterval, q0, w0, eps, qLo, qHi, wLo, wHi,
          scale, powers, termLo, termHi, k, value1, value2, i,
j,
          box, kappaBox, r, finiteBound, eta1, eta2,
cauchyBound,
          displayScale, kappaLower, kappaUpper, finiteUpper,
          cauchyUpper, totalUpper;

    q0 := 3092493386004/10^13;
    w0 := 2320376944290/10^12;
    eps := 1/10^12;
    qLo, qHi := q0-eps, q0+eps;
    wLo, wHi := w0-eps, w0+eps;
    if qLo <= 0 or wLo <= 0 or qHi+1/200 >= 1/3 or wHi+1/2 >= 3
then
        error "the Cauchy domain is outside the stated bounds";
    end if;

    scale := 10^60;
    # The coefficient tail in (33) is < 1/scale for i <= 19 and
j <= 9.
    if 2^(19+2*9+1) >= 3^40 or 2*41^19*820^9*scale >= 3^779 then
        error "the coefficient tail estimate fails";
    end if;

    powers := Array(0..40);
    termLo := Array(0..40);
    termHi := Array(0..40);
    for k from 0 to 40 do
        powers[k] := k*(k-1)/2;
        value1 := (-1)^k*scale*wLo^k*qLo^(powers[k]);
        value2 := (-1)^k*scale*wHi^k*qHi^(powers[k]);
        termLo[k] := floor(min(value1, value2));
        termHi[k] := ceil(max(value1, value2));
    end do;

    # Enclose C[i,j] using the signed term intervals.
    CoefficientInterval := proc(ii, jj, e, lowerTerms,
upperTerms, grid)
        local index, multiplier, lowerSum, upperSum;
        # Enclose the omitted tail after scaling.
        lowerSum, upperSum := -1, 1;
        for index from ii to 40 do
            if e[index] >= jj then
                multiplier := binomial(index,
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ii)*binomial(e[index], jj);
                lowerSum := lowerSum +
multiplier*lowerTerms[index];
                upperSum := upperSum +
multiplier*upperTerms[index];
            end if;
        end do;
        return [lowerSum/grid, upperSum/grid];
    end proc;

    kappaBox := CoefficientInterval(2, 0, powers, termLo,
termHi, scale);
    r := 3/50;
    finiteBound := 0;
    for i from 0 to 19 do
        for j from 0 to 9 do
            if i+2*j >= 3 then
                box := CoefficientInterval(i, j, powers, termLo,
termHi, scale);
                finiteBound := finiteBound
                    + max(abs(box[1]),
abs(box[2]))*(3/2)^i*r^(i+2*j-2);
            end if;
        end do;
    end do;

    # The terms through k = 4 sum to 8 + 1/9.
    # From k = 5 the first term is 1/243 and the ratio is <=
1/81.
    if 8+1/9+(1/243)/(1-1/81) >= 10 then
        error "the Cauchy majorant is too large";
    end if;
    eta1 := 3*wHi*r;
    eta2 := 200*qHi*r^2;
    if eta1 >= 1 or eta2 >= 1 then
        error "the Cauchy ratios are not below one";
    end if;
    cauchyBound := 10/r^2*(eta1^20+eta2^10)/((1-eta1)*(1-eta2));

    # Round the displayed interval endpoints and upper bounds
outward.
    displayScale := 10^14;
    kappaLower := floor(displayScale*kappaBox[1]);
    kappaUpper := ceil(displayScale*kappaBox[2]);
    finiteUpper := ceil(displayScale*finiteBound);
    cauchyUpper := ceil(displayScale*cauchyBound);
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    totalUpper := ceil(displayScale*(finiteBound+cauchyBound));

    if kappaLower/displayScale < 7033861652/10^10 or
       kappaUpper/displayScale > 7033861654/10^10 then
        error "the printed kappa enclosure fails";
    end if;
    if finiteUpper/displayScale >= 118838/10^6 or
       cauchyUpper/displayScale >= 2/1000 or
       totalUpper/displayScale >= 120838/10^6 or 120838/10^6 >=
1/8 then
        error "a printed remainder bound fails";
    end if;

    printf("kappa in [0.%014d, 0.%014d]   PASS\n", kappaLower,
kappaUpper);
    printf("Finite Taylor contribution <= 0.%014d < 0.118838
PASS\n", finiteUpper);
    printf("Cauchy remainder <= 0.%014d < 0.002   PASS\n",
cauchyUpper);
    printf("Total remainder <= 0.%014d < 0.120838 < 1/8
PASS\n", totalUpper);
end proc:

VerifyLocalTaylorBounds():
kappa in [0.70338616529259, 0.70338616533741]   PASS
Finite Taylor contribution <= 0.11883706829264 < 0.118838   PASS
Cauchy remainder <= 0.00199808027206 < 0.002   PASS
Total remainder <= 0.12083514856469 < 0.120838 < 1/8   PASS
>
```